

Nanoelectronics and Technology
Critical Comprehensive Practice Paper (50 Questions)
Marking Rubric and Model Answers (v2)

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Part I: Examination Paper (50 Questions)

Instructions

Answer in coherent paragraphs; use equations when they sharpen the argument. For “figure” questions, assume the described plot is provided. Keep answers within the syllabus scope: lithography (EBL, NIL, DUV/EUV), memory (SRAM/DRAM/Flash/RRAM/MRAM+STT), microscopy (STM/AFM/SEM/TEM+BF/DF/EELS), and reliability (yield, traps, SILC/TDDB, HK/IL, grain boundaries, injection).

Section A: Lithography (Q1–Q12)

1. Pattern transfer pipeline: coat $\rightarrow$ bake $\rightarrow$ expose $\rightarrow$ PEB $\rightarrow$ develop $\rightarrow$ etch/lift-off $\rightarrow$ strip. Where are CD, overlay, and LER primarily set, and why?
2. Positive vs negative resist: chemistry-level difference, what dissolves, and consequences for (i) lift-off and (ii) etch masking.
3. Resolution scaling: interpret $CD \approx k_1 \lambda / NA$. What does k_1 hide, why does low k_1 increase stochastic risk, and what happens to yield if CD halves but overlay error stays fixed?

4. DUV immersion: use $NA = n \sin \theta$ to explain why immersion can give $NA > 1$. Name two immersion-specific defect mechanisms and how they impact yield.
5. Multiple patterning vs EUV: compare mask count, overlay stacking, variability, and cost. Give one scenario where DUV multipatterning is still preferable.
6. EUV optics: why are lenses impossible at 13.5 nm? Why mirrors? If each mirror has reflectivity R , and there are N reflections, derive intensity scaling and link it to throughput.
7. EUV vacuum + contamination: why vacuum is mandatory, and why contamination still happens. Give two mitigation strategies and their tradeoffs.
8. EUV source (tin LPP): why tin, what limits conversion efficiency, and why higher source power reduces stochastic failures.
9. **Figure:** LER decreases with dose then saturates; throughput decreases with dose. Explain the curve shapes and how to pick an operating point.
10. EBL proximity: forward vs backscattering, why high- Z substrates worsen proximity, and how PEC (dose modulation/fracturing) fixes it.
11. Why not mass-manufacture with EBL? Provide (i) throughput argument, (ii) materials/charging/resist argument.
12. NIL: why it can beat diffraction yet struggles with overlay, why defects replicate, and why residual layer etch is required.

Section B: Memory Devices (Q13–Q30)

13. Unify “a stored bit”: identify the physical state variable for SRAM, DRAM, flash, RRAM, MRAM; define volatility using the same language.
14. SRAM bistability and SNM: explain cross-coupled inverter latch and SNM (butterfly concept). Which sizing ratios improve read SNM vs write margin?
15. SRAM write: for writing $Q = 1$, specify $BL, \overline{BL}, WL$ and which devices must overpower which. What fails if access is too weak? if pull-ups are too strong?
16. SRAM read and read disturb: explain precharge, differential discharge, sensing. Why can a read flip the cell?
17. **Figure:** two butterfly curves with different lobe size. Interpret and give two variation sources that shrink SNM.
18. DRAM charge sharing: derive ΔV for C_c at V_{DD} sharing with C_{BL} at $V_{DD}/2$; explain why ΔV is small.
19. DRAM destructive read + restore: explain why read destroys and why sense amp restores; relate to row-level operation.
20. DRAM refresh: “how does it know the bit” without a second memory? Explain the full row refresh timeline and leakage link.

21. Flash floating gate electrostatics: explain V_T shift via capacitive coupling; define program/erase in terms of electron motion and V_T direction.
22. FN vs direct tunneling vs HCI: why FN barrier is triangular; compare fields, locality, and damage.
23. Flash erase: why it is harsher; connect to endurance via trap creation and widening V_T distributions.
24. RRAM basis: why TMOs switch; distinguish VCM vs ECM in one coherent paragraph.
25. Forming vs set vs reset: why forming is riskier and one-time. What catastrophic outcomes occur if forming overshoots?
26. Bipolar vs unipolar: polarity reversal vs Joule heating; role of compliance current.
27. RRAM variability: why filaments are stochastic; what if multiple filaments form; two strategies to reduce variability.
28. MTJ stack: pinned/free layers, MgO barrier, SAF/exchange bias; what stores the bit and what ensures non-volatility?
29. TMR: explain why parallel is lower resistance; include spin polarization and MgO symmetry filtering.
30. MRAM read: why small bias; what is read disturb and how is it avoided?
31. STT write: explain spin-transfer torque; give qualitative scaling of I_c with anisotropy/volume/damping; retention vs write-energy tradeoff.
32. Engineering comparison: SRAM/DRAM/flash/RRAM/STT-MRAM on speed, endurance, density, leakage, integration; propose a plausible hierarchy.

Section C: Microscopy and Characterization (Q31–Q43)

31. STM: constant-current imaging; derive/justify $I \propto e^{-2\kappa d}$; why STM maps LDOS, not pure height.
32. AFM modes: contact vs non-contact vs tapping; what observable is used (deflection/amplitude/ Δf) and one artifact per mode.
33. FM-AFM: derive $\Delta f = -\frac{f_0}{2k} \frac{\partial F}{\partial z}$ and interpret the sign/meaning of force gradient.
34. SEM signals: SE vs BSE generation and escape depth; why SE = topography, BSE = Z contrast; kV tradeoff.
35. **Figure:** one SEM image highlights edges, another highlights heavy inclusions. Identify SE vs BSE and justify.
36. SEM resolution: why interaction volume matters; how lowering kV and field-emission sources change resolution and contrast.
37. TEM basics: why transmission requires thin specimens; why FIB prep; what damage FIB introduces and mitigations.

38. TEM planes and apertures: image plane vs diffraction (back focal) plane; why aperture selection is Fourier filtering, not magic.
39. BF TEM: which beam(s) selected; mass-thickness vs diffraction contrast; how dislocations show up.
40. DF TEM: select a diffracted beam; why grains/precipitates light up; how to use DF for orientation/phase mapping.
41. BF vs DF twisted why: why BF still has diffraction contrast, and why DF can reveal defects invisible in BF.
42. EELS: low-loss vs core-loss; plasmons vs inner-shell excitations; what ELNES tells you.
43. **Figure:** EELS spectrum with ZLP, plasmon peak, and a core edge. Identify each and explain thickness effects.

Section D: Reliability, Yield, High- k /IL (Q44–Q50)

44. Yield: derive Poisson $Y = e^{-D_0 A}$; why clustering breaks Poisson and what qualitative model change is needed.
45. SILC/TDDB/soft vs hard breakdown: define each; how SILC arises; what soft vs hard implies for circuits.
46. Traps: define properly; explain trap-assisted tunneling and why traps “enable” conduction without being metallic wires.
47. Hydrogen release model + forming gas anneal: passivation (Si–H), stress-induced breakage, interface traps, relation to BTI.
48. Thermochemical model + anode hole injection: field-lowered activation, give a correct exponential rate form; what AHI is and why holes are damaging.
49. HK/IL, grain boundaries, substrate injection, catastrophic failure: why IL exists, EOT/interface tradeoff, grain boundary leakage, substrate injection mechanism, and percolation-to-breakdown picture.

Part II: Marking Rubric

Global rule: mechanism + “why”

Full credit answers must (i) state the governing mechanism, (ii) connect cause $\rightarrow$ effect, (iii) include at least one correct scaling/equation where relevant, and (iv) include one meaningful limitation/tradeoff.

Default marking per question (10 marks)

- **2** Correct definition/scope + addresses all subparts.
- **4** Core mechanism with correct causality and sign logic.
- **2** One correct relation/scaling/selection argument (e.g., $CD \sim \lambda/NA$, $I \sim R^N$, $I \sim e^{-2\kappa d}$, charge sharing).
- **2** Tradeoff/failure mode + practical implication.

Figure questions (10 marks)

- **3** Identify what is plotted and the controlling variable(s).
- **4** Explain trend using correct physics (noise, scattering, charge sharing, etc.).
- **2** Operational decision or diagnosis (how a process engineer would act).
- **1** Confounder/artifact and a check.

High-penalty conceptual errors

- EUV “lenses” or air operation; ignoring R^N reflectivity cascade.
- SEM vs TEM confusion (emission vs transmission; thick vs thin requirement).
- Treating DF TEM as a label instead of “diffracted-beam-selected imaging”.
- Traps as “conductive pieces”; ignoring TAT/percolation.
- Flash as generally “write-once” (it is rewritable, endurance-limited).

Part III: Model Answers

Section A Answers

1. **A1.** CD is primarily set by the aerial image + resist response: exposure dose/focus and the resist/process (PEB and development) convert the optical/e-beam intensity into an edge position. Overlay is set by alignment metrology + stage/optics + cumulative layer-to-layer registration, and it manifests when the pattern is placed relative to previous layers. LER is “born” from stochasticity and material granularity: photon/electron shot noise, acid generation/diffusion (for CAR), polymer/molecular discreteness, and developer kinetics. Post steps (etch bias, etch roughening) can amplify LER, but its root is the latent-image-to-edge randomness.
2. **A2.** Positive resist: exposure increases solubility (e.g., chain scission or deprotection), so exposed regions dissolve. Negative resist: exposure decreases solubility by crosslinking, so exposed regions remain. Lift-off prefers negative or bilayer positive strategies because you want resist profiles with undercut to prevent metal “fencing”; crosslinked negatives often give robust lift-off profiles. For etch masking, positive resists are common for high-resolution openings; the key is etch selectivity and profile control, not ideology. Negative resists can be mechanically tougher but may suffer swelling and worse LER at small CDs depending on chemistry.
3. **A3.** k_1 hides illumination coherence, mask bias, phase-shift tricks, OPC/RET, and process window engineering. Very low k_1 means you are printing near the limit where small dose/focus variations shift edges strongly, and where stochastic effects dominate because fewer photons/electrons define the edge. If CD halves but overlay stays fixed, overlay becomes a larger fraction of feature size, so misalignment causes shorts/opens more often: yield drops sharply because overlay error budget does not scale automatically with CD.
4. **A4.** Since $NA = n \sin \theta$, putting a fluid with $n > 1$ between lens and wafer increases NA beyond 1 while still respecting $\sin \theta \leq 1$. This increases the spatial frequencies captured, improving resolution. Immersion worsens defectivity (bubbles/particles print as defects) and introduces fluid/resist interactions (leaching, swelling), both shrinking process window and harming yield. It also adds complex topcoat requirements and tighter control of fluid handling and temperature uniformity.
5. **A5.** Multiple patterning reduces effective pitch by splitting one dense layout into multiple less-dense exposures/etches, but it increases mask count, process steps, and overlay stacking: every split adds alignment and CD variability that compounds. EUV reduces the number of splits for some critical layers, lowering overlay accumulation and steps, but introduces EUV-specific stochastic defects and high tool cost/complexity. DUV multipatterning can still be preferable when EUV stochastic defectivity is unacceptable for a layer, when EUV tool availability is limited, or when the layer’s pitch can be met with mature DUV + spacers at lower risk/cost.
6. **A6.** At 13.5 nm, materials absorb strongly, so lenses would attenuate the beam and heat up; refractive optics are infeasible. Multilayer mirrors exploit Bragg reflection to achieve usable reflectivity at EUV. If each mirror has reflectivity R , after N reflections the intensity scales as $I = I_0 R^N$. Since dose on wafer must meet a target for resist, lower I means longer exposure time, directly lowering throughput (wafers/hour). This is why mirror reflectivity and mirror count are existential in EUV.

7. **A7.** Vacuum is mandatory because air absorbs EUV over millimeters; you would lose essentially all intensity and create uncontrolled chemistry. Contamination persists because EUV cracks residual hydrocarbons and resist outgassing, depositing carbon films on mirrors (reflectivity loss), and because tin debris from the source can reach optics. Mitigation: hydrogen background can chemically clean carbon (tradeoff: can affect resist/process and requires careful control), and debris mitigation (magnetic fields, foil traps) protects optics (tradeoff: reduces collected EUV and adds complexity). Also, low-outgassing resists reduce contamination (tradeoff: may worsen sensitivity or LER).
8. **A8.** Tin is used because the plasma emission spectrum has strong intensity near 13.5 nm and practical droplet generation. Conversion efficiency is limited by how laser energy partitions into EUV vs other wavelengths/heat, plasma self-absorption, and collection geometry; debris mitigation also reduces usable flux. Higher source power reduces stochastic failures by allowing higher on-wafer dose at a fixed throughput or maintaining dose while increasing throughput; more photons reduce shot-noise-driven stochastic effects (missing/bridging at tiny features).
9. **A9.** At low dose, photon/electron shot noise dominates edge placement; increasing dose increases the number of quanta and reduces relative noise, lowering LER. Saturation occurs when other floors dominate: acid diffusion blur, polymer granularity, development noise, and etch-induced roughening. Manufacturing picks an operating point by minimizing total cost of defects + time: choose the smallest dose that meets LER/defectivity spec with acceptable throughput, often using a “knee” in the curve where LER improvements per extra dose become marginal.
10. **A10.** Forward scattering broadens the beam as electrons penetrate resist, smearing exposure laterally. Backscattered electrons from the substrate travel laterally before re-entering resist, creating a long-range exposure tail (proximity). High- Z substrates increase backscatter coefficient, worsening the long-range dose halo and making dense/isolated features print differently. PEC works by predicting the deposited energy convolution and pre-distorting the pattern/dose: increase dose where energy is “stolen” and reduce it where halo would overexpose, often via feature fracturing and local dose modulation.
11. **A11.** Throughput: EBL is serial. A chip requires enormous shot count; even with high beam current you hit space-charge blur, resist heating, and limited resist sensitivity. Projection lithography is parallel: a whole field is exposed at once. Materials: charging on insulating substrates deflects the beam; secondary electron blur limits ultimate edge fidelity; resist sensitivity vs resolution tradeoff forces slow writes for fine features. Hence EBL is unbeatable for masks/R&D, not for HVM wafers.
12. **A12.** NIL can beat diffraction because the mold physically defines nanoscale relief, not an aerial image. Overlay is hard because imprint is contact: mechanical distortion (thermal expansion, wafer bow, resist flow) makes alignment nontrivial across the field. Defects replicate because the mold is reused; a particle on mold or wafer prints a defect repeatedly and can damage the mold. Residual layer etch is needed because after imprint there is typically a thin resist film at the bottom of trenches; anisotropic etch clears it to open features for transfer. This step affects CD because over-etch can widen features and adds variability.

Section B Answers

13. **B13.** SRAM state = a pair of node voltages $(Q, \overline{Q})$ held by positive feedback; volatility means power loss collapses the latch and state disappears. DRAM state = charge (voltage) on a capacitor relative to reference; volatility means leakage discharges it with time even if power stays. Flash state = trapped charge on floating gate shifting V_T ; non-volatility means barrier prevents charge loss without applied bias. RRAM state = resistance (LRS/HRS) set by defect/ion configuration (filament); non-volatility means that configuration is metastable at zero bias. MRAM state = free-layer magnetization direction relative to reference; non-volatility means anisotropy energy barriers prevent spontaneous flipping.
14. **B14.** Two cross-coupled inverters create regenerative feedback: a small increase in Q drives $\overline{Q}$ lower, which further drives Q higher, leading to one of two stable rails. SNM is the maximum DC noise that can be injected at the nodes without flipping the latch; butterfly curves quantify it via the largest inscribed square between the two inverter VTCs. Read SNM improves with strong pull-down NMOS (cell ratio) so the “0” node is not lifted during read. Write margin improves with strong access transistors relative to pull-ups so bitlines can force a node across the trip point. These compete: stronger pull-down helps read but can hinder write of “1” depending on sizing, hence ratio optimization.
15. **B15.** To write $Q = 1$, typically force $BL = 1$, $\overline{BL} = 0$, assert WL to turn on access NMOS. The $\overline{BL} = 0$ side must pull the target node that should become 0 below the inverter trip point, at which point feedback flips the cell, making the opposite node rise to 1. If access is too weak, bitlines cannot overcome the existing latch, so the write fails (cell does not flip). If pull-ups are too strong, the “0”-forcing bitline cannot discharge the node enough; again write fails. If pull-downs are too strong relative to access, you can also get contention that slows or prevents flipping.
16. **B16.** Read: precharge BL and $\overline{BL}$ high, then assert WL . The node storing “0” provides a discharge path: bitline $\rightarrow$ access NMOS $\rightarrow$ cell pull-down NMOS $\rightarrow$ ground, so that bitline drops slightly; the other bitline stays high. A sense amp detects the small differential and amplifies it. Read disturb occurs because the “0” node is connected to a precharged high bitline through the access NMOS, creating charge sharing that raises the node voltage. If that node rises past the inverter trip point, feedback flips the cell. Prevent by sizing (strong pull-down, controlled WL , lower precharge, assist circuits).
17. **B17.** Larger butterfly lobes mean larger SNM (more tolerance to noise and variation). Smaller lobes imply higher susceptibility to read upset and retention failure. SNM shrinks due to (i) threshold mismatch and mobility variation (random dopants, LER, stress), which distorts inverter VTCs, and (ii) supply droop/IR drop or increased leakage, which lowers gain and compresses VTCs. Process corners, temperature, and local layout effects (line width variation) also reduce lobe area.
18. **B18.** Charge conservation: $C_c V_{DD} + C_{BL}(V_{DD}/2) = (C_c + C_{BL})V_f$. Thus $V_f = \frac{C_c V_{DD} + C_{BL} V_{DD}/2}{C_c + C_{BL}}$ and $\Delta V = V_f - V_{DD}/2 = \frac{C_c}{C_c + C_{BL}} \cdot \frac{V_{DD}}{2}$. Since typically $C_{BL} \gg C_c$, ΔV is small, so thermal noise and device mismatch can overwhelm it unless a strong regenerative sense amplifier converts the tiny differential into full logic rails quickly.
19. **B19.** Read is destructive because connecting C_c to the large C_{BL} forces the cell voltage toward the bitline precharge level through charge sharing, erasing the original stored level. The sense

amp is effectively a latch connected to complementary bitlines; once enabled, it amplifies the sign of ΔV and drives one bitline to V_{DD} and the other to 0. Because WL is still high during sensing, the cell capacitor is recharged/discharged to match the sensed bit, restoring the state. This is why DRAM read and restore are inseparable at the circuit level.

20. **B20.** Refresh is row-wise periodic read-and-restore. The system “knows” the bit because the cell itself perturbs the precharged bitline up or down; the sense amp detects that perturbation and then forces the full rail values, rewriting the cell while resolving the bitline. Timeline: precharge to $V_{DD}/2$, raise WL for a row, allow charge sharing to create $\pm\Delta V$, enable sense amps to latch and drive rails, then drop WL and restore precharge. Leakage (junction leakage, subthreshold, dielectric leakage) sets the retention time, hence refresh interval.
21. **B21.** Floating-gate charge shifts V_T because the FG is capacitively coupled to control gate and channel. Negative charge on FG screens the control-gate field, requiring higher V_G to invert the channel, so V_T increases (nMOS cell). Program typically injects electrons onto FG (raising V_T); erase removes electrons (lowering V_T). Read senses whether channel conducts at a given wordline voltage by measuring current, so the memory is effectively stored as a V_T distribution rather than a literal “voltage on a node”.
22. **B22.** FN tunneling happens under high oxide field that tilts the barrier into a triangular shape, thinning it and enabling tunneling through a field-lowered barrier; it is strongly field-dependent. Direct tunneling occurs when the oxide is thin enough that electrons tunnel through a trapezoidal barrier even at moderate fields; it scales violently with thickness. Hot-carrier injection uses large lateral field near drain to accelerate carriers; some surmount the barrier into oxide (non-tunneling or assisted) and get trapped. FN typically stresses a larger area/oxide path; HCI is localized near drain, often causing localized damage and different endurance behavior.
23. **B23.** Erase is harsher because it often requires larger opposite-polarity fields and longer times, and may involve tunneling paths that drive carriers through interfaces that are more damage-prone. Program/erase cycling creates traps in tunnel oxide and at interfaces, which increases leakage and causes V_T drift and distribution widening, eventually leading to read errors and reduced endurance. Even before catastrophic breakdown, trap accumulation makes the cell less distinguishable, raising raw bit error rate.
24. **B24.** TMOs switch because their defect chemistry allows creation/migration of oxygen vacancies and redox changes that strongly modulate conductivity. In VCM, switching is vacancy-driven: set forms a vacancy-rich conductive filament/region, reset oxidizes or disrupts it, changing resistance. In ECM, switching is metal-ion-driven: an active electrode dissolves (Ag/Cu), ions migrate and plate to form a metallic filament (set), and reverse bias dissolves it (reset). Thus VCM is oxygen/stoichiometry modulation; ECM is electrochemical metal filament formation/dissolution.
25. **B25.** Forming is needed because pristine oxide lacks a conductive pathway; a higher voltage creates an initial defect network/filament that makes subsequent switching possible at lower voltages. Set then repeatedly forms/thickens filament to LRS; reset ruptures/thins to HRS. Forming overshoot can cause permanent short (hard breakdown), uncontrolled multi-filament formation, electrode damage, or thermal runaway that changes the device into an irreversible conductor, i.e., “dead cell”.

26. **B26.** Bipolar: set and reset require opposite polarities because ionic drift direction must reverse (vacancies/ions move one way to form, opposite to dissolve). Unipolar: same polarity; reset is mainly Joule-heating-driven rupture at controlled power/current. Compliance current limits the maximum current during set/form so that filament formation does not run away into a permanent short; it is the difference between reversible switching and killing the device.
27. **B27.** Filaments form where local defect density and temperature favor percolation, which is inherently stochastic at nanoscale. Multiple filaments increase conductance and can create partial-reset behavior, complicating binary separation and multi-level control; they also spread stress unpredictably, affecting endurance. Variability reduction: (i) confine switching to a designed region (nano-oxide, engineered constriction, interface layer) to force filament nucleation at a controlled site; (ii) use material engineering (dopants, oxygen reservoir layers, electrode selection) to stabilize vacancy distributions; also verify with tighter compliance and adaptive programming pulses.
28. **B28.** MTJ: reference/pinned ferromagnet (often SAF + exchange bias) / MgO tunnel barrier / free ferromagnet. Bit is stored as relative magnetization alignment of free vs pinned (P or AP). Non-volatility comes from magnetic anisotropy energy barriers that make the free layer stable in either orientation at zero current/field; thermal agitation is insufficient to flip it over the barrier for the desired retention time.
29. **B29.** $TMR = (R_{AP} - R_P)/R_P$ quantifies conductance contrast. Parallel is lower resistance because spin-polarized electrons in the majority channel of one electrode tunnel efficiently into available majority states of the other when magnetizations align; in AP alignment, a majority spin on one side corresponds to minority on the other, reducing transmission. Crystalline MgO enhances TMR by symmetry filtering: certain Bloch state symmetries couple efficiently through MgO, so alignment strongly modulates the dominant tunneling channel, increasing contrast beyond an amorphous barrier.
30. **B30.** Read applies small bias to measure MTJ resistance without exceeding switching threshold; it is ideally non-destructive because magnetization stays put. Read disturb is unintended switching caused by read current producing spin torque near I_c (or by heating). Avoid by limiting read current/pulse width, increasing I_c margin, and using read circuits that sense quickly with minimal charge/time.
31. **B31.** STT writing: a spin-polarized current carries angular momentum; when it enters the free layer, it exerts torque that can overcome damping and anisotropy to reverse magnetization. Critical current scales with free-layer volume V , damping α , and anisotropy field/energy H_k (qualitatively $I_c \propto \alpha V H_k$ divided by spin polarization efficiency). Higher anisotropy and larger volume improve retention (higher barrier) but increase write current/energy; lowering them eases write but harms retention and increases error probability.
32. **B32.** SRAM: fastest, high endurance, but low density and leakage. DRAM: high density and cost-effective, but refresh and destructive read; moderate speed. Flash: non-volatile, ultra-dense, low leakage; slow writes/erases and endurance-limited. RRAM: non-volatile, potentially dense and fast; variability/forming/endurance issues. STT-MRAM: non-volatile, fast-ish, high endurance; write current and integration complexity. Hierarchy: SRAM for caches, DRAM for main memory, flash for storage; STT-MRAM as embedded non-volatile cache/working memory for instant-on; RRAM as potential future storage-class memory if variability is tamed.

Section C Answers

31. **C31.** Tunneling current depends on barrier width because electron wavefunction decays in the classically forbidden gap: $\psi \sim e^{-\kappa d}$, so transmission $T \sim e^{-2\kappa d}$ and $I \propto T$. Constant-current mode uses feedback to adjust tip height so I stays fixed as the tip scans, producing a height map. STM maps LDOS because $I(V)$ depends on the energy-resolved density of states of tip and sample near the Fermi level; at fixed bias, variations in electronic structure (surface states, bandgap regions, adsorbates) change I even at the same geometric height, so the image is an electronic-topographic hybrid.
32. **C32.** Contact AFM measures static deflection proportional to force; artifact: frictional drag and sample damage, especially on soft materials. Non-contact AFM measures frequency shift from force gradient; artifact: requires stable environment, and long-range forces (electrostatics, capillary) can dominate and blur true topography. Tapping AFM measures oscillation amplitude/phase changes with intermittent contact; artifact: can still damage delicate features if setpoint too aggressive, but greatly reduces lateral shear compared to contact, making it the practical default for heterogeneous and soft samples.
33. **C33.** With $f_0 = \frac{1}{2\pi} \sqrt{k/m}$, a conservative interaction adds an effective stiffness change $\Delta k = -\partial F/\partial z$, so $f = \frac{1}{2\pi} \sqrt{(k + \Delta k)/m} \approx f_0 (1 + \frac{\Delta k}{2k}) = f_0 (1 - \frac{1}{2k} \frac{\partial F}{\partial z})$. Therefore $\Delta f \approx -\frac{f_0}{2k} \frac{\partial F}{\partial z}$. The sign means attractive gradients ($\partial F/\partial z > 0$ for increasing attraction as you approach) reduce effective stiffness and lower frequency; repulsive gradients increase stiffness and raise frequency. Force gradient is literally “how fast the force changes with distance,” which is what the oscillator is sensitive to.
34. **C34.** SE are low-energy electrons produced by inelastic scattering near the surface; low escape depth makes them topography-sensitive because geometry controls escape probability. BSE are incident electrons elastically scattered back; deeper origin and Z -dependent scattering gives compositional contrast. Raising kV increases penetration and interaction volume, boosting BSE and X-ray yield but reducing surface sensitivity and increasing charging/damage for insulators; lowering kV improves surface sensitivity and reduces damage but reduces signal and may increase noise.
35. **C35.** Edge-bright image is SE because edges enhance SE escape and yield; heavy-inclusion-bright image is BSE because high- Z regions backscatter more. The underlying physics is inelastic near-surface emission for SE vs elastic Z -dependent scattering for BSE.
36. **C36.** Even with a tiny beam spot, electrons scatter inside the sample; the region where signals are generated (interaction volume) sets the effective resolution, especially for BSE and X-rays. Lowering kV shrinks interaction volume and makes SE more surface-specific; field-emission sources reduce source size and improve brightness/coherence, enabling smaller probe sizes and better resolution at low kV . Tradeoff: low kV reduces BSE contrast and can increase charging sensitivity depending on sample.
37. **C37.** TEM requires electrons to transmit with limited multiple scattering, hence thin specimens (tens of nm). FIB allows site-specific thinning of device cross-sections with nanoscale placement. Damage: Ga implantation, amorphization, redeposition, curtaining, heating. Mitigation: low-kV final polish, protective caps (Pt), cleaning cross-sections, cryo or gentle ion milling, and careful beam conditions to reduce implanted dose and amorphous layer thickness.

38. **C38.** The objective lens produces a real-space image plane and a back focal (diffraction) plane where scattering angle maps to position; this is a Fourier transform relationship between object and spatial frequencies. An aperture in the diffraction plane selects which angular components contribute, implementing spatial-frequency filtering. BF centers the aperture on the transmitted beam; DF centers it on a diffracted spot. This is not magic: it is the same optics principle that a lens maps angles to positions and separates frequency components, enabling controlled contrast mechanisms.
39. **C39.** BF TEM selects the transmitted beam (and possibly low-angle) with the objective aperture. Mass-thickness contrast: thicker/heavier regions scatter more electrons out of the transmitted beam, appearing darker. Diffraction contrast: regions satisfying Bragg conditions redirect intensity into diffracted beams, also reducing transmitted intensity, producing contrast linked to orientation/strain. Dislocations strain the lattice locally, shifting Bragg condition and scattering, giving characteristic line contrast in BF images depending on diffraction condition.
40. **C40.** DF TEM selects a specific diffracted beam; only regions that strongly contribute to that diffraction spot appear bright. Therefore grains oriented to satisfy the chosen Bragg condition light up, while others remain dark. For mapping orientation, you can take DF images while selecting different spots; different grains appear bright in different selections, giving orientation contrast. For phase identification, select diffraction spots unique to a phase or lattice spacing; precipitates of that phase will brighten selectively.
41. **C41.** BF still shows diffraction contrast because diffraction removes intensity from the transmitted beam: even if diffracted beams are blocked, the transmitted beam intensity varies with local Bragg satisfaction and dynamical scattering. DF can reveal defects invisible in BF because selecting a diffracted beam can amplify contrast from a specific distortion/phase that weakly affects transmitted intensity but strongly affects that diffracted channel. In other words, BF is “transmitted-beam depletion” contrast; DF is “selected-scattered-beam enrichment” contrast.
42. **C42.** Low-loss EELS measures near-elastic and valence excitations: zero-loss peak and plasmons, giving thickness and dielectric response. Core-loss EELS measures inner-shell ionization edges, giving element-specific composition. ELNES (near-edge fine structure) reflects unoccupied density of states and local bonding/oxidation state, so EELS can probe chemistry and coordination. Because inelastic scattering is angle- and thickness-dependent, acquisition uses spectrometers and collection angles chosen to balance signal and plural scattering.
43. **C43.** ZLP corresponds to electrons that did not lose measurable energy (elastic transmission). The plasmon peak is collective valence-electron excitation; it broadens and grows with thickness due to more inelastic events and plural scattering. The sharp core edge corresponds to an inner-shell excitation threshold of a specific element; its onset identifies the element, and fine structure reveals bonding. As thickness increases, plural scattering increases background and can obscure fine structure; deconvolution or thinner specimens are used for accurate ELNES.

Section D Answers

44. **D44.** If defects are random with areal density D_0 , expected defects on a die of area A is $\lambda = D_0 A$. Under Poisson statistics, yield (probability of zero killer defects) is $Y = P(0) = e^{-\lambda} = e^{-D_0 A}$. Clustering breaks independence: defects come in bursts, so the zero-defect probability is higher than Poisson for small areas and worse for large areas depending on clustering parameters.

Qualitatively, you need a model with non-Poisson variance (e.g., negative binomial or Murphy-type) to account for defect clustering.

45. **D45.** SILC is leakage increase after stress due to newly created traps that enable trap-assisted tunneling/hopping through the dielectric. TDDB is time-dependent dielectric breakdown: progressive defect accumulation under field leading to breakdown at time-to-failure. Soft breakdown means a partial conductive path forms, increasing leakage but not necessarily shorting the gate completely; circuits may still function with higher static power or altered bias. Hard (catastrophic) breakdown is a strong percolation path causing near-short and immediate functional failure or severe leakage that collapses node voltages.
46. **D46.** A trap is a localized electronic state in the bandgap created by a structural defect (vacancy, dangling bond, impurity, distorted bond). It can capture and emit carriers, altering charge and enabling conduction. In trap-assisted tunneling, an electron tunnels electrode $\rightarrow$ trap and then trap $\rightarrow$ other electrode; multiple traps form a stepping-stone path that reduces the effective tunneling distance per step, dramatically increasing leakage compared to direct tunneling through a pristine oxide. Traps are not metallic wires; they are quantum states that mediate carrier transport and charge trapping.
47. **D47.** Forming gas anneal (H_2/N_2) passivates Si dangling bonds at Si/SiO₂ (or IL) by forming Si-H, reducing interface trap density and improving mobility and threshold stability. Under electric field and temperature, Si-H bonds can break; hydrogen is released and dangling bonds reappear as interface traps (D_{it} rises). Mobile hydrogen can diffuse and create additional defects. This hydrogen release model underlies bias temperature instability (BTI) by generating and activating interface/oxide traps, shifting V_T over time.
48. **D48.** Thermochemical model: defect creation is a field- and temperature-accelerated bond rupture process with rate $\Gamma \propto \exp\left[-\frac{E_a - \alpha E}{kT}\right]$ (one common form), where field lowers effective activation energy and higher T increases reaction rate. Anode hole injection (AHI) occurs under strong electron injection (e.g., FN) when energetic electrons reaching the anode generate electron-hole pairs; holes are injected into the dielectric from the anode side. Holes are damaging because they can break bonds and create positively charged traps and are strongly coupled to defect creation pathways in many oxides.
49. **D49.** High- k /IL: an interfacial SiO_x layer improves interface quality on Si (lower D_{it} , better mobility) but increases EOT because SiO_x has low k ; HfO₂ reduces EOT for a given physical thickness but can crystallize and introduce defects. Grain boundaries in polycrystalline high- k provide defect-rich leakage pathways and local field enhancement, increasing leakage and accelerating breakdown. Substrate injection refers to carriers injected from the Si substrate/channel into the dielectric under high field (direct/FN/hot carriers), which creates traps and increases SILC/TDDB risk. Catastrophic failure arises when trap density reaches a percolation threshold, forming a continuous conductive path (hard breakdown) that collapses gate insulation; high- k stacks face extra challenges due to grain boundaries, interfacial complexity, and different defect chemistry compared to SiO₂.